

Project Assignment Report: Traffic Signal Synchronization System

Project Title: Traffic Signal Synchronization System Using Chinese Remainder Theorem

Course / Assignment: Number Theory & Discrete Mathematics Application

Team Leadership: Justin Joe Mathew (Team Lead)

Does Traffic Signal Synchronization Demo CRT or just LCM/GCD?

Mathematical Distinction:

1. LCM (Least Common Multiple) / GCD:

Used when all signals start at Minute 0 with NO remainder/offset. If Signal 1 (3 min), Signal 2 (5 min), and Signal 3 (7 min) all turn green at Minute 0, the next simultaneous green light is simply $\text{LCM}(3, 5, 7) = 105$ minutes.

2. CRT (Chinese Remainder Theorem):

REQUIRED when signals have DIFFERENT REMAINDERS / OFFSETS.

In our project scenario:

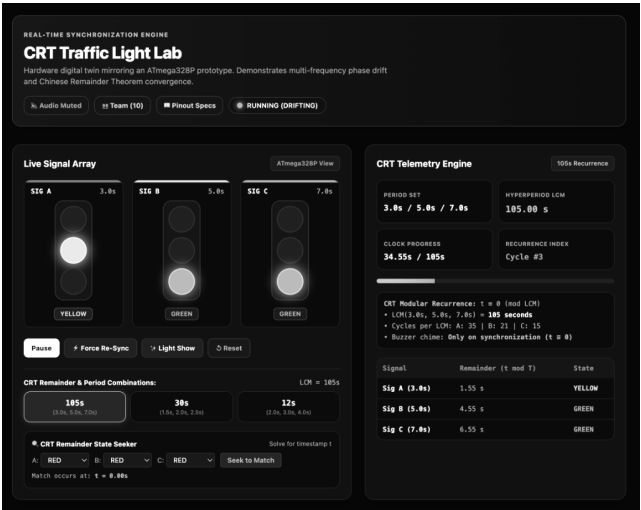
- Signal 1 (3-min cycle) turned green 2 minutes ago $\rightarrow x \equiv 2 \pmod 3$
- Signal 2 (5-min cycle) turned green 3 minutes ago $\rightarrow x \equiv 3 \pmod 5$
- Signal 3 (7-min cycle) turned green 2 minutes ago $\rightarrow x \equiv 2 \pmod 7$

Because the remainders (2, 3, 2) are non-zero and distinct, LCM cannot solve this problem. The system requires the Chinese Remainder Theorem (Extended Euclidean Algorithm & Modular Inverses) to solve for the unique green wave minute ($x = 23$).

Conclusion: The project is a genuine demonstration of CRT, specifically showcasing why CRT is needed over simple LCM when real-world cycles start at different offsets.

Physical Hardware Prototype & Digital Twin Interface

The project includes both a physical Arduino Nano breadboard circuit and a full digital twin web simulator.



Multi-Channel Phase Waveform

White cursor sweeps across the 105s hyperperiod. Cyan vertical line marks simultaneous alignment.

Window: 0.00s - 185.00s

SIG A

SIG B

SIG C

Arduino Nano Physical Firmware

arduino/traffic_light_controller.ino

Copy Arduino Code

Signal A: Pin 2 (Red), Pin 3 (Yellow), Pin 4 (Green) • Signal B: Pin 8 (Red), Pin 9 (Yellow), Pin 18 (Green) • Signal C: Pin A5 (Red), Pin A4 (Yellow), Pin A3 (Green) • Buzzer: Pin A1

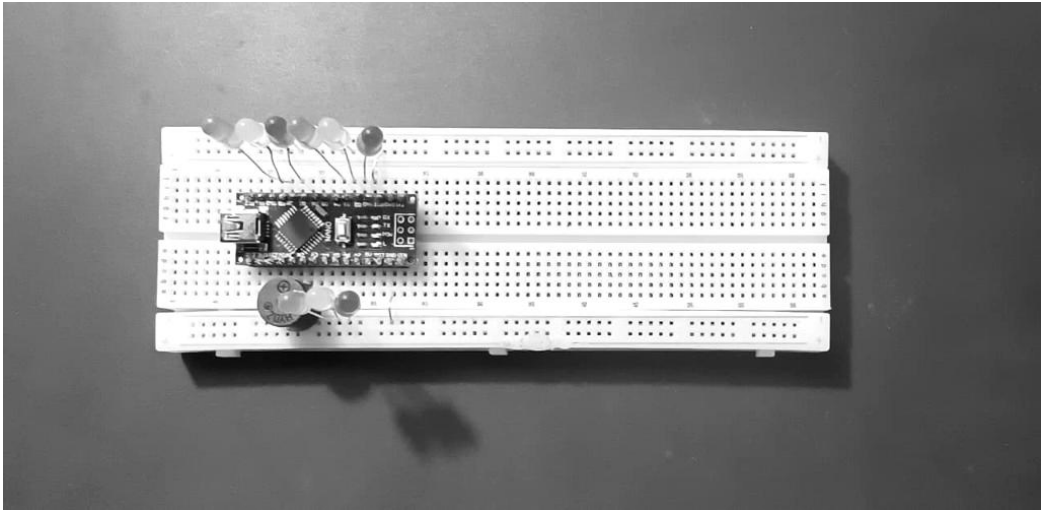
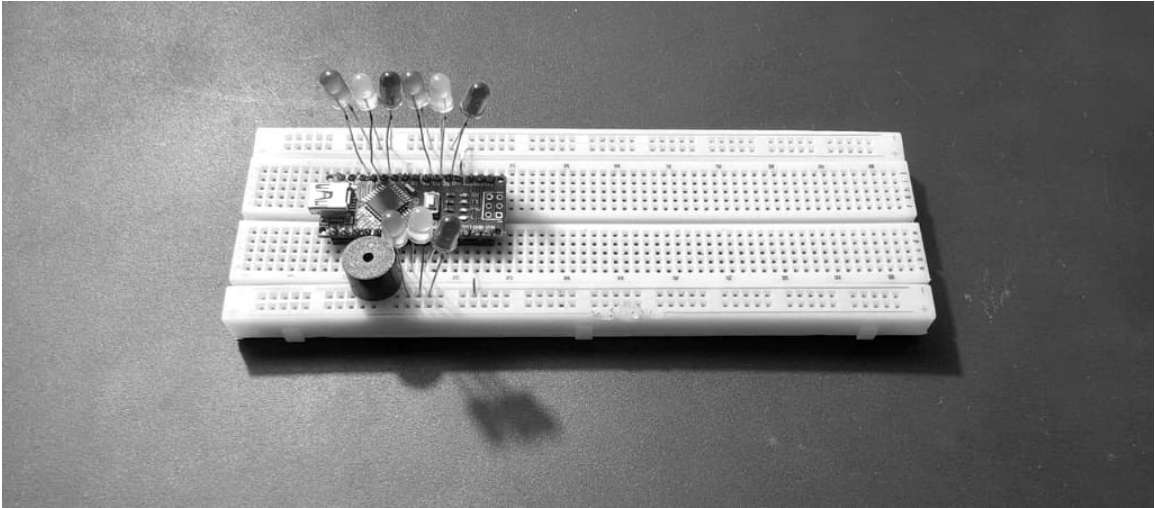
```
const int RED_A = 2;
const int YELLOW_A = 3;
const int GREEN_A = 4;

const int RED_B = 8;
const int YELLOW_B = 9;
const int GREEN_B = 18;

const int GREEN_C = A3;
const int YELLOW_C = A4;
const int RED_C = A5;

const int BUZZER = A1;

const unsigned long PERIOD_A = 3000;
const unsigned long PERIOD_B = 5000;
const unsigned long PERIOD_C = 7000;
const unsigned long MASTER_CYCLE = 185000;
```



■ Group Member Roles & Work Division (10 Members)

PROJECT TEAM STRUCTURE		
■ 1. Project Leadership & Model	■ Justin Joe Mathew (Team Lead) & Jyothis Liju	■
■ 2. Hardware Sourcing & Build Testing	■ Joseph Alex & Joseph J	■
■ 3. Web Design & Frontend	■ Johan Geo & Johaan Sam	■
■ 4. Presentation & Showcase	■ Joel Geo Manuel	■
■ 5. Mathematics & Applications	■ Joel Duke, Jyothika Prakash, & Jose Alex	■

■ Detailed Team Member Contribution Breakdown

1. Project Leadership & Working Model Construction

■ Team Members: Justin Joe Mathew (Team Lead) & Jyothis Liju

- Justin Joe Mathew (Team Lead): Guided overall project direction, engineered core system architecture, and led the construction and logic of the working traffic signal model.
- Jyothis Liju: Assisted Justin directly in constructing the working traffic signal model, testing signal synchronization alignment, and verifying physical component interaction.

2. Hardware Sourcing, Build Testing & Application Design

■ Team Members: Joseph Alex & Joseph J

- Joseph Alex: Determined component requirements, purchased hardware parts, executed physical build testing, and contributed to application use-cases.
- Joseph J: Handled hardware part selection, procurement, mechanism assembly testing, and co-developed application integration ideas.

3. Web Design & Frontend Development

■ Team Members: Johan Geo & Johaan Sam

- Johan Geo: Designed the user interface, visual layout, traffic signal dashboard theme, and interactive monitor dials.
- Johaan Sam: Built the frontend components, state integration, responsive layouts, and interactive controls.

4. Presentation & Demo Lead

■ Team Member: Joel Geo Manuel

- Joel Geo Manuel: Handled the presentation structure, demo walkthrough flow, visual slides, and showcase delivery for explaining the project to an audience.

5. Mathematics & CRT Applications Research

■ Team Members: Joel Duke, Jyothika Prakash, & Jose Alex

- Joel Duke: Researched linear congruences and formulated the traffic signal synchronization modulo equations (\$3, 5, 7\$ minute cycles).
- Jyothika Prakash: Explored real-world CRT applications and authored step-by-step mathematical explanations and hint structures.
- Jose Alex: Investigated Chinese Remainder Theorem proofs, solution bounds (\$N = 105\$), and modulo remainder properties.